

Lecture 8

Orthonormal Bases, Gram–Schmidt, and Riesz

Inner products are in the physics convention (conjugate-linear in the first slot). Tags mark the flavour; a ★ marks a harder one.

Orthonormal bases and Gram–Schmidt

Exercise 8.1 (computational) Apply Gram–Schmidt to $v_1 = (1, 1, 1)$, $v_2 = (1, 1, 0)$, $v_3 = (1, 0, 0)$ in $\mathbb{R}^3$ with the dot product.

Exercise 8.2 (computational) Expand $v = (2, 3) \in \mathbb{R}^2$ in the orthonormal basis $e_1 = \frac{1}{\sqrt{2}}(1, 1)$, $e_2 = \frac{1}{\sqrt{2}}(1, -1)$, and verify Parseval's relation.

Exercise 8.3 (computational) Apply Gram–Schmidt to $v_1 = (3, 4)$, $v_2 = (1, 0)$ in $\mathbb{R}^2$ with the dot product.

Exercise 8.4 (computational) Expand $v = (3, 0, 0) \in \mathbb{R}^3$ in the orthonormal basis $e_1 = \frac{1}{3}(1, 2, 2)$, $e_2 = \frac{1}{3}(2, 1, -2)$, $e_3 = \frac{1}{3}(2, -2, 1)$, and check Parseval's relation.

Exercise 8.5 (computational) Orthonormalize $v_1 = (1, i)$, $v_2 = (0, 1)$ in $\mathbb{C}^2$ (physics convention, conjugate-linear in the first slot).

Exercise 8.6 (proof) Prove that an orthonormal list $e_1, \dots, e_m$ is linearly independent directly from the definition.

Exercise 8.7 (proof) Let $e_1, \dots, e_n$ be an orthonormal basis of V . Prove that for all $u, v \in V$,

$$\langle u, v \rangle = \sum_{k=1}^n \overline{\langle e_k, u \rangle} \langle e_k, v \rangle,$$

so the coordinate map $v \mapsto (\langle e_1, v \rangle, \dots, \langle e_n, v \rangle)$ preserves inner products.

Projections and best approximation

Exercise 8.8 (computational) Find the point of the line $U = \text{span}((1, 2, 2)) \subset \mathbb{R}^3$ closest to $b = (3, 0, 0)$.

Exercise 8.9 (computational) Fit a line $y = c_0 + c_1 x$ to the data $(1, 1)$, $(2, 2)$, $(3, 2)$ by least squares.

Exercise 8.10 (computational) Find the orthogonal projection of $b = (2, 4) \in \mathbb{R}^2$ onto the line $U = \text{span}((1, 1))$, and the distance from b to U .

Exercise 8.11 (computational) Project $b = (3, 0, 0) \in \mathbb{R}^3$ onto the plane U with orthonormal basis $e_1 = \frac{1}{\sqrt{2}}(1, 1, 0)$, $e_2 = \frac{1}{\sqrt{6}}(1, -1, 2)$.

Exercise 8.12 (computational) Solve the inconsistent system $Ax = b$ in the least-squares sense, where $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}$ and $b = (1, 1, 0)$.

Exercise 8.13 (proof) Suppose $P = P_U$ is an orthogonal projection. Prove $P^2 = P$ and that $v - Pv \perp Pv$ for every v .

Exercise 8.14 (proof) Let P_U be the orthogonal projection onto a subspace U with orthonormal basis $e_1, \dots, e_m$. Prove that $\langle P_U v, w \rangle = \langle v, P_U w \rangle$ for all $v, w \in V$.

Exercise 8.15 (★) On $\mathcal{P}_2$ with $\langle p, q \rangle = \int_{-1}^1 \bar{p} q dx$, find the polynomial of degree ≤ 1 closest to $f(x) = x^2$, and verify that the error is orthogonal to $\mathcal{P}_1$.

Riesz and orthogonal complements

Exercise 8.16 (computational) Find the vector $w \in \mathbb{C}^2$ representing the functional $\varphi(v) = 2v_1 - iv_2$ in the form $\varphi(v) = \langle w, v \rangle$.

Exercise 8.17 (computational) For $U = \text{span}((1, 0, 1)) \subset \mathbb{R}^3$, describe $U^\perp$ and verify $\dim U + \dim U^\perp = 3$.

Exercise 8.18 (computational) Find $w \in \mathbb{R}^3$ representing the functional $\varphi(v) = 2v_1 - v_2 + 3v_3$ in the form $\varphi(v) = \langle w, v \rangle$.

Exercise 8.19 (computational) Find $w \in \mathbb{C}^3$ with $\varphi(v) = \langle w, v \rangle$ for $\varphi(v) = iv_1 + 2v_2 - (1+i)v_3$.

Exercise 8.20 (computational) For $U = \text{span}((1, 0, 1, 0), (0, 1, 0, 1)) \subset \mathbb{R}^4$, give a basis of $U^\perp$ and check $\dim U + \dim U^\perp = 4$.

Exercise 8.21 (computational) On $\mathcal{M}_2(\mathbb{R})$ with the Frobenius product $\langle A, B \rangle = \text{tr}(A^\dagger B)$, find the matrix w representing $\varphi(A) = 2A_{11} - A_{12} + A_{22}$ as $\varphi(A) = \langle w, A \rangle$.

Exercise 8.22 (★) Let $e_1, \dots, e_n$ be an orthonormal basis and $T \in \mathcal{L}(V)$. Prove that $\text{tr}(T^\dagger T) = \sum_{j,k} |\langle e_j, T e_k \rangle|^2 \geq 0$, with equality iff $T = 0$.

Exercise 8.23 (★) Suppose φ is a linear functional on V represented by w via $\varphi(v) = \langle w, v \rangle$. Prove that $\text{null } \varphi = \{w\}^\perp$ and, if $w \neq 0$, that $\dim \text{null } \varphi = \dim V - 1$.

Exercise 8.24 (proof) Let U be a subspace of an inner product space V . Prove directly from the definitions that $U \subseteq (U^\perp)^\perp$.